Abstract Linear Algebra

Prof. Lanier
jlanier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

Next up:

- This week: Chapter 3
- Quizzes 4 & 5 today
- Problem session today, 3-5pm, Ryerson 177.
- Homework 3 is due tonight / tomorrow.
- HW4 distributed today, due end of next week.
- ***Change:*** HW3 debrief is moved to next Wednesday 4-5pm, Eckhart 207A
- No OH Sunday
- Requizzing: Monday 3:30-4:30 in Eckhart 207A, Quizzes 1-5

MATH 20250
Prof. Lanier

Midterm Review Problems

April 14, 2023

The following problems will help you to prepare for the midterm exam. They indicate the style (but not the amount!) of problems that will form the bulk of the midterm: 1. giving examples, 2. correcting important definitions, 3. stating and illustrating important results, 4. proving smaller results, 5. ASN, 6-10. free response questions, mostly similar in content, style, and difficulty to the homework. I've emphasized content here from Ch 1 & 2, since you'll be working on many Ch 3 problems in the next week. A key for these review problems will be posted and problems will be discussed at the review session on Sunday, April 23.

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1. Give an example of each of the following, or explain why it is not possible.

a) A matrix that is similar to an identity matrix but not row equivalent to it.

b) A matrix that is row equivalent to an identity matrix but not similar to it.

c) a 3×7 matrix with nullity 4

d) a 2×5 matrix with rank 3

e) A basis for $\mathbb{C}^3$ that contains no standard basis vectors

f) A span in $\mathbb{C}^3$ that contains no standard basis vectors

g) A vector space with a pair of subspaces that are disjoint

h) Four different examples of elementary matrices

i) a pair of matrices that are matrix equivalent but not row equivalent

j) a pair of matrices that are row equivalent but not matrix equivalent

k) A non-identity matrix A such that $A^2 = A^{-1}$.

l) A linear transformation with domain $\mathbb{R}^3$ and codomain $\mathbb{C}^2$

m) a pair of similar matrices with different determinants

n) a pair of matrices of different dimensions but the same determinant

o) three real 2×2 matrices with integer entries whose inverses also have integer entries

p) A matrix with nullspace equal to its column space

q) A matrix with nullity equal to its rank

r) five different fields

s) a pair of 2×2 matrices that have the same determinant but are not similar

t) a vector space with exactly two subspaces

u) a vector space with exactly three subspaces

v) a vector space with exactly five subspaces

w) a matrix that is upper triangular but not diagonal

x) a matrix that is both symmetric and antisymmetric

y) a linear transformation that is neither 1-to-1 nor onto

z) a map between two vector spaces that is not a linear transformation

Ten Important Terms

A **vector space** is a set V of vectors with the structure of a commutative group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

A **linear combination** of vectors is a finite sum of scalar multiples of the vectors. For vectors $v_1, v_2, \ldots, v_p$, a linear combination of is a sum of form $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_p v_p = \Sigma_1^p \alpha_k v_k$ where the α_k are scalars. The **span** of a set of vectors in a vector space is the set of all of their linear combinations.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and scalar multiplication: $L(\alpha v + \beta w) = \alpha L(v) + \beta L(w)$.

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a unique representation as a linear combination of the vectors in the basis. The cardinality of a basis is the **dimension** of a vector space. (*Note: it is a theorem that dimension is well-defined.*)

A matrix is in **row echelon form** if all zero rows (if any) are below all non-zero rows, and if every non-zero row has its leading non-zero entry strictly to the right of the first non-zero entry in the previous row; the leading non-zero entry in each non-zero row is called its **pivot**. A matrix is in **reduced row echelon form** if it also has a 1 for each of its pivots and all entries above each pivot are 0.

The **kernel** of a linear transformation $L : V \to W$ is the subset of V sent to the zero vector in W by L. The **range** of L is the subset of W in the image of L. (*It is a result that the kernel and range are subspaces of V and W, respectively; they are also called the* **null space** *and* **column space**).

Two matrices A and B are **similar** (or **conjugate**) if there exists an invertible matrix Q such that $A = Q^{-1}BQ$.

For the square matrices $M_{n\times n}$ over a field $\mathbb{F}$, a **determinant** is a function $\det : M_{n\times n} \to \mathbb{F}$ such that det is multilinear and alternating and $\det(I) = 1$, where I is the $n \times n$ identity matrix. (*Note: it is a theorem that a determinant exists and is unique.*)

Ten Important Results: Chapters 1-3

Any finite generating system for a vector space contains a basis. Every basis of a vector space has the same cardinality, yielding a well-defined dimension. (Prop. 1.2.8; Prop. 2.3.3)

Among the set of matrices that are row equivalent to a given matrix, there exists a unique matrix in reduced row echelon form. (Exercise 2.3.9; see handout.)

Linear Algebra Terms (for midterm)

Chapters 1 and 2:
vector
scalar
group
field
matrix
vector space
linear transformation
subspace
transpose
linear combination
trivial (linear combination, solution, subspace)
linear independence (and dependence)
span (generate)
spanning system
basis
standard basis
coordinates (in a basis)
dimension
isomorphism
rank
column space (range)
null space (kernel)
row space
one-to-one (injective)
onto (surjective)
isomorphism
conjugation, change of basis, similarity
homogeneous and inhomogeneous systems
vector equation
matrix equation
inconsistent and consistent systems
unique solution
trivial solution
general solution
coefficient matrix
augmented matrix
row operations
row equivalent
row reduction
row echelon form
reduced row echelon form (RREF)
pivot
free variable
(parametric) vector form
fundamental subspaces of a matrix
matrix equivalent
types of matrices/linear transformations:
zero
identity
elementary
projection
scaling
shear
rotation
permutation
diagonal
upper triangular
lower triangular
invertible (nonsingular)
non-invertible (singular)
symmetric (equal to transpose)
Chapter 3:
determinant
parallelpiped
submatrix
minor of order k
cofactor expansion
multilinear
antisymmetric or alternating
normalized

HW3

III. Always sometimes, or never? (If sometimes, give an example each way. If always or never, explain why.)

i) If A and B are square matrices and $AB = B$, then A is an identity matrix.

ii) The rank of a matrix M is equal the rank of M^T.

iii) The nullity of a matrix M is equal to the nullity of M^T.

iv) A square matrix is similar to an identity matrix.

v) A square matrix is row equivalent to an identity matrix.

An invertible vi) ~~A square~~ matrix is similar to an identity matrix.

vii) An invertible matrix is row equivalent to an identity matrix.

Three methods for computing matrix inverses (when they exist):

- use geometric intuition
- use row reduction
- use a formula

Five methods for computing the determinant of a matrix:

- use geometric intuition
- use column (or row) reduction
- use a formula
- notice a special form
- use cofactor expansion

Column operations on a matrix affect its determinant in predictable ways.

scaling a column by k scales the determinant by k

adding a multiple of a column to a given column preserves the determinant

swapping two columns negates the determinant

the identity matrix has determinant 1

Special forms:

- The determinant of a diagonal matrix is the product of its diagonal entries.
- The determinant of an upper (or lower) triangular matrix is the product of its diagonal entries
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More generally, if a square matrix has linearly dependent columns, its determinant is 0.

A invertible $\Longleftrightarrow$ A (square) has linearly independent columns

$\Updownarrow$

$\det(A) \neq 0$

Another equivalent condition of the invertible matrix theorem

Cofactor expansion of determinant (Thm 3.5.1) For an $n \times n$ matrix A, let $A_{j,k}$ denote the $(n-1) \times (n-1)$ matrix obtained from A by crossing out row number j and column number k. For any fixed j, we have

$$\det A = \sum_{k=1}^{n} a_{j,k}(-1)^{j+k} \det A_{j,k}.$$

Cofactor expansion is a way to compute the determinant of a matrix
by computing determinants of several smaller sub-matrices,
weighting them appropriately,
and then summing up the results.

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = \underline{\qquad\qquad} + \underline{\qquad\qquad} + \underline{\qquad\qquad}$$

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$$\begin{vmatrix} \boxed{1 \quad 2 \quad 3} \\ 4 \quad 5 \quad 6 \\ 7 \quad 8 \quad 9 \end{vmatrix} = 1 \cdot {+1} \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + 2 \cdot {-1} \cdot \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot {+1} \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

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$$\begin{vmatrix} \boxed{1 \quad 2 \quad 3} \\ 4 \quad 5 \quad 6 \\ 7 \quad 8 \quad 9 \end{vmatrix} = \underline{1 \quad +1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix}} + \underline{2 \quad -1 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix}} + \underline{3 \quad +1 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}}$$

$$= -3 + 12 - 9 = 0$$

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} \qquad \begin{vmatrix} 1 & \cancel{2} & \cancel{3} \\ \cancel{4} & 5 & 6 \\ \cancel{7} & 8 & 9 \end{vmatrix} \qquad \begin{vmatrix} \cancel{1} & 2 & \cancel{3} \\ 4 & \cancel{5} & 6 \\ 7 & \cancel{8} & 9 \end{vmatrix} \qquad \begin{vmatrix} \cancel{1} & \cancel{2} & 3 \\ 4 & 5 & \cancel{6} \\ 7 & 8 & \cancel{9} \end{vmatrix}$$

Now we calculate the same determinant using row reduction:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \underset{R2-4R1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{vmatrix} \underset{R3-7R1}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{vmatrix} \underset{R3-2R2}{=} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

matrices where it’s easy to compute the determinant:

- diagonal matrices
- upper and lower triangular matrices
- special case: elementary matrices
- matrices with lots of zero entries

the determinant

There exists a unique map $M_{n\times n}^{\mathbb{F}} \to \mathbb{F}$ that is multilinear, antisymmetric and normalized, which is the determinant.

(Section 3.4)

linear in columns (and rows)

the determinant

There exists a unique map $M^{\mathbb{F}}_{n\times n} \to \mathbb{F}$ that is multilinear, antisymmetric and normalized, which is the determinant.

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$$L(av) = aL(v) \qquad L(v+w) = L(v) + L(w)$$

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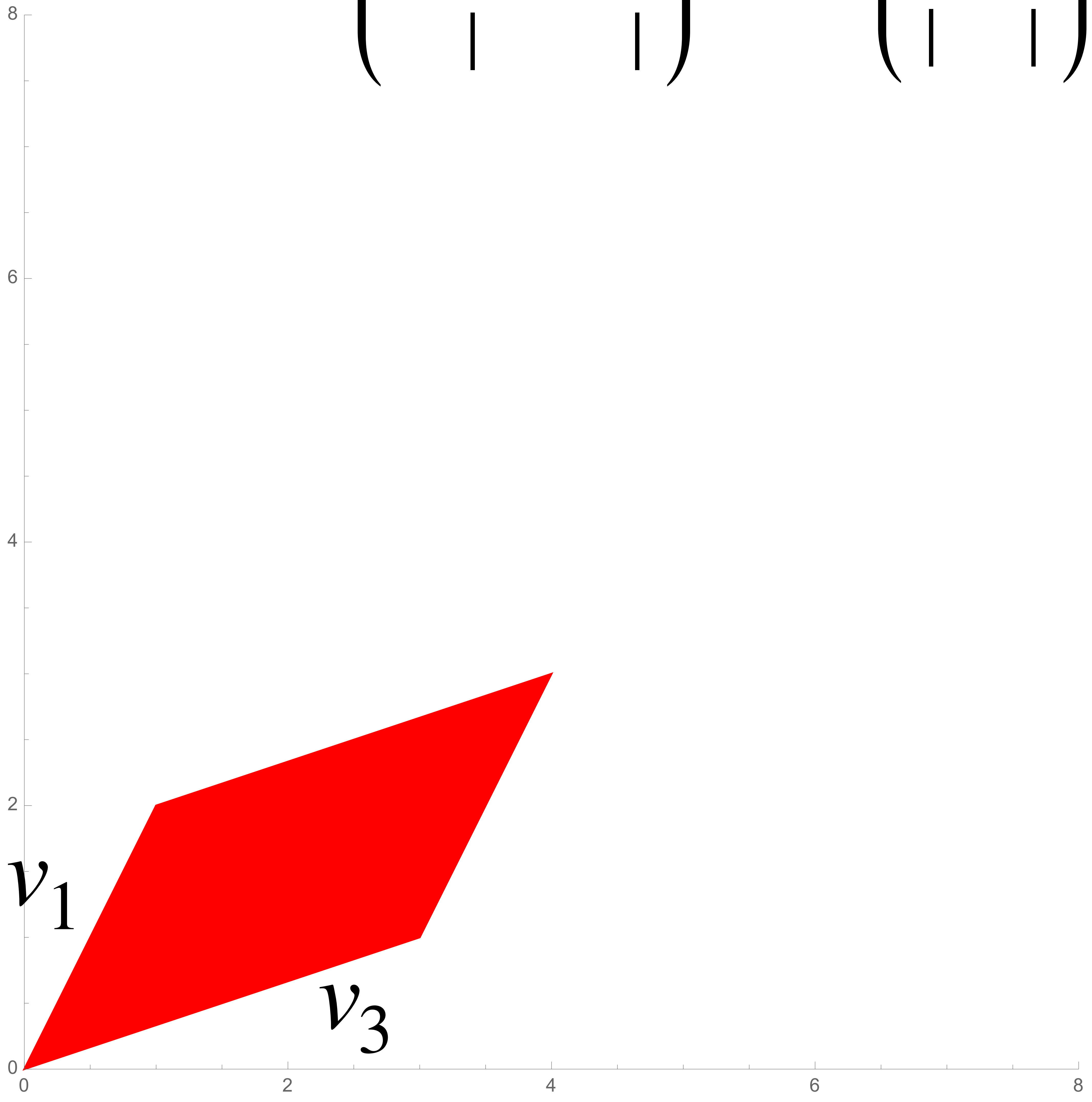
$$\det\begin{pmatrix} | & | & | \\ av_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} = a\det\begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}$$

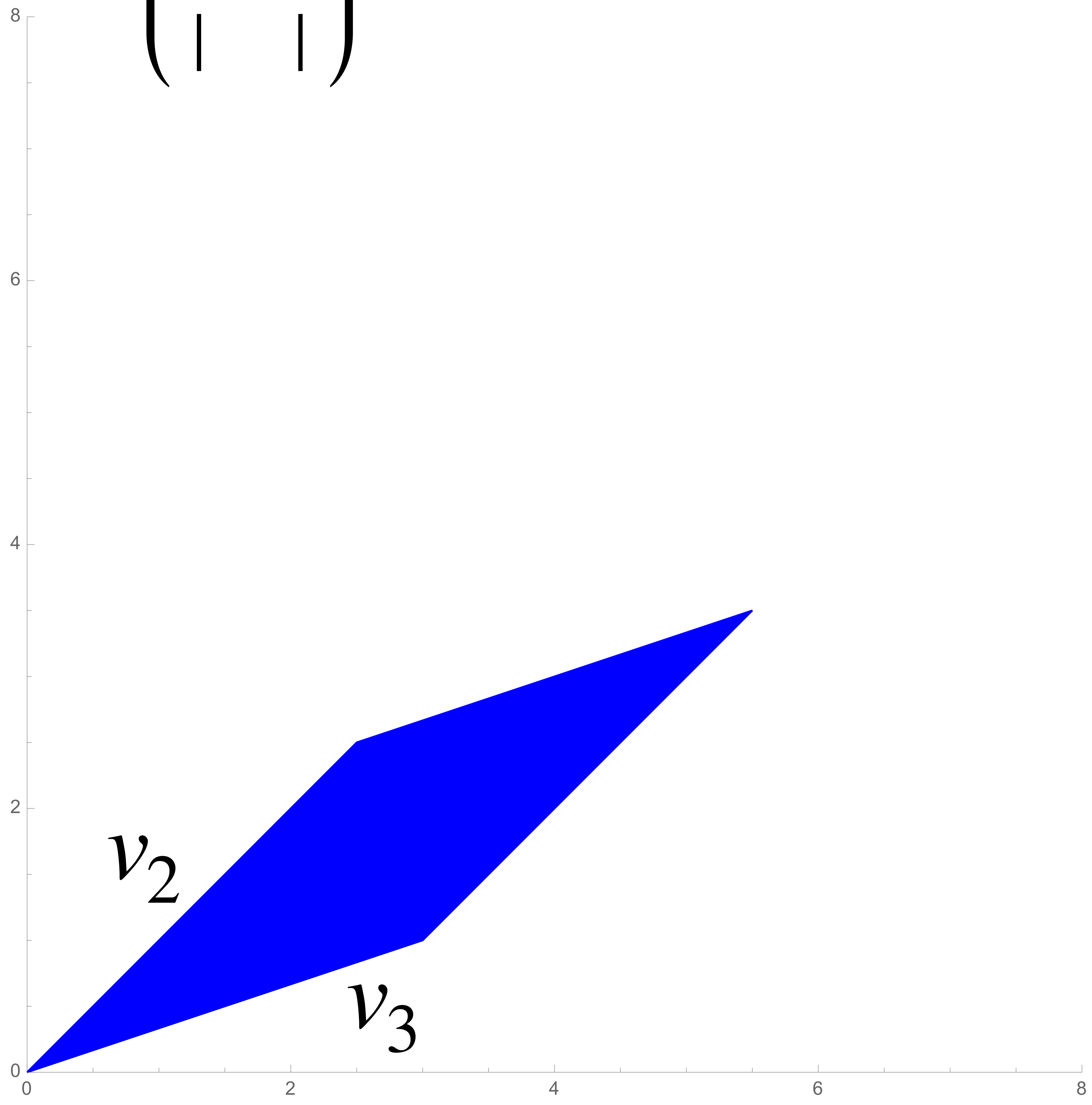
$$\det\begin{pmatrix} | & | & | \\ v_1+v_4 & v_2 & v_3 \\ | & | & | \end{pmatrix} = \det\begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} + \det\begin{pmatrix} | & | & | \\ v_4 & v_2 & v_3 \\ | & | & | \end{pmatrix}$$

Volume scales linearly when stretched in one direction.

Volume adds nicely when you have a common base.

$$\det\begin{pmatrix} | & | \\ v_1 + v_2 & v_3 \\ | & | \end{pmatrix} = \det\begin{pmatrix} | & | \\ v_1 & v_3 \\ | & | \end{pmatrix} + \det\begin{pmatrix} | & | \\ v_2 & v_3 \\ | & | \end{pmatrix}$$





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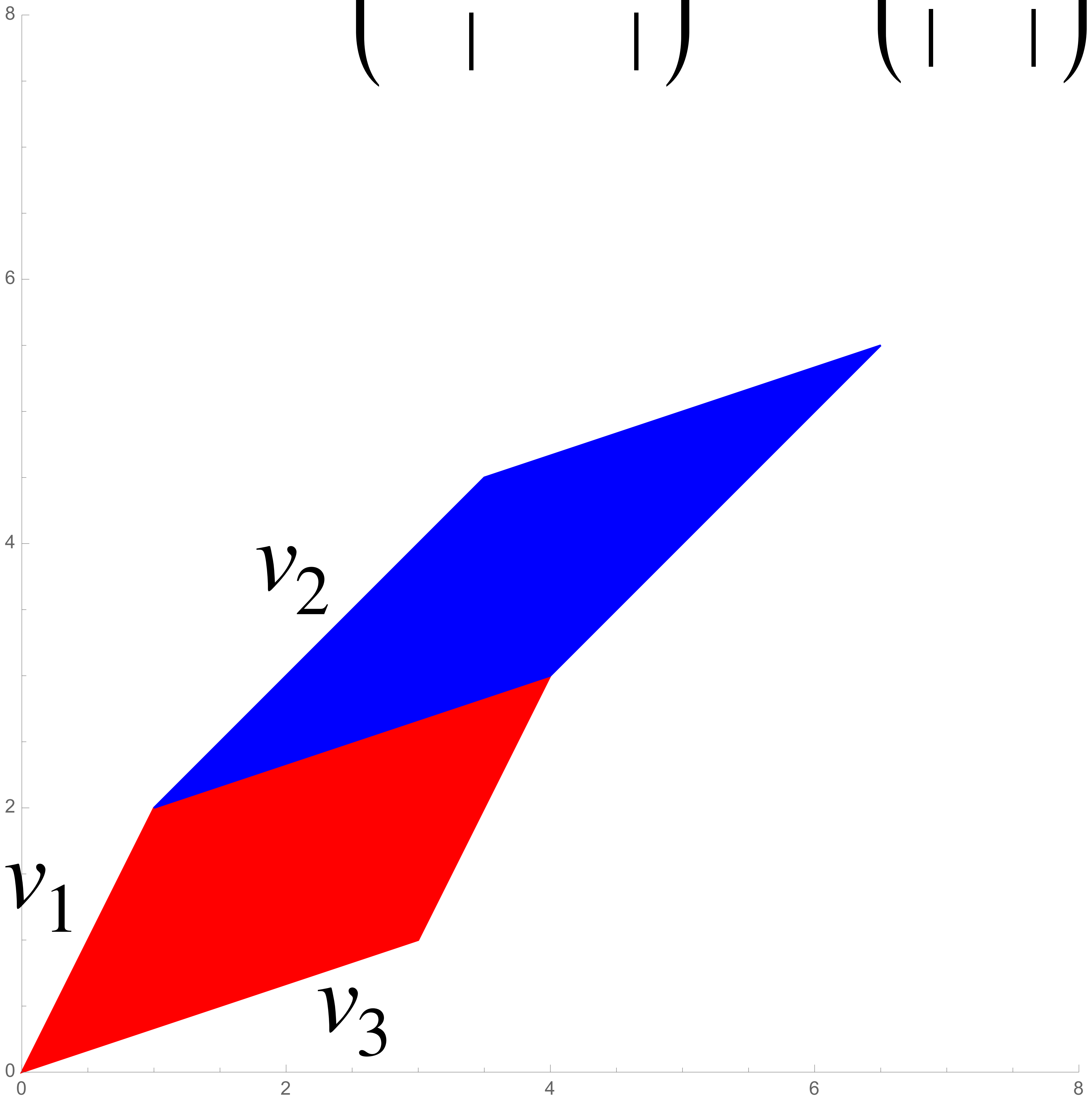

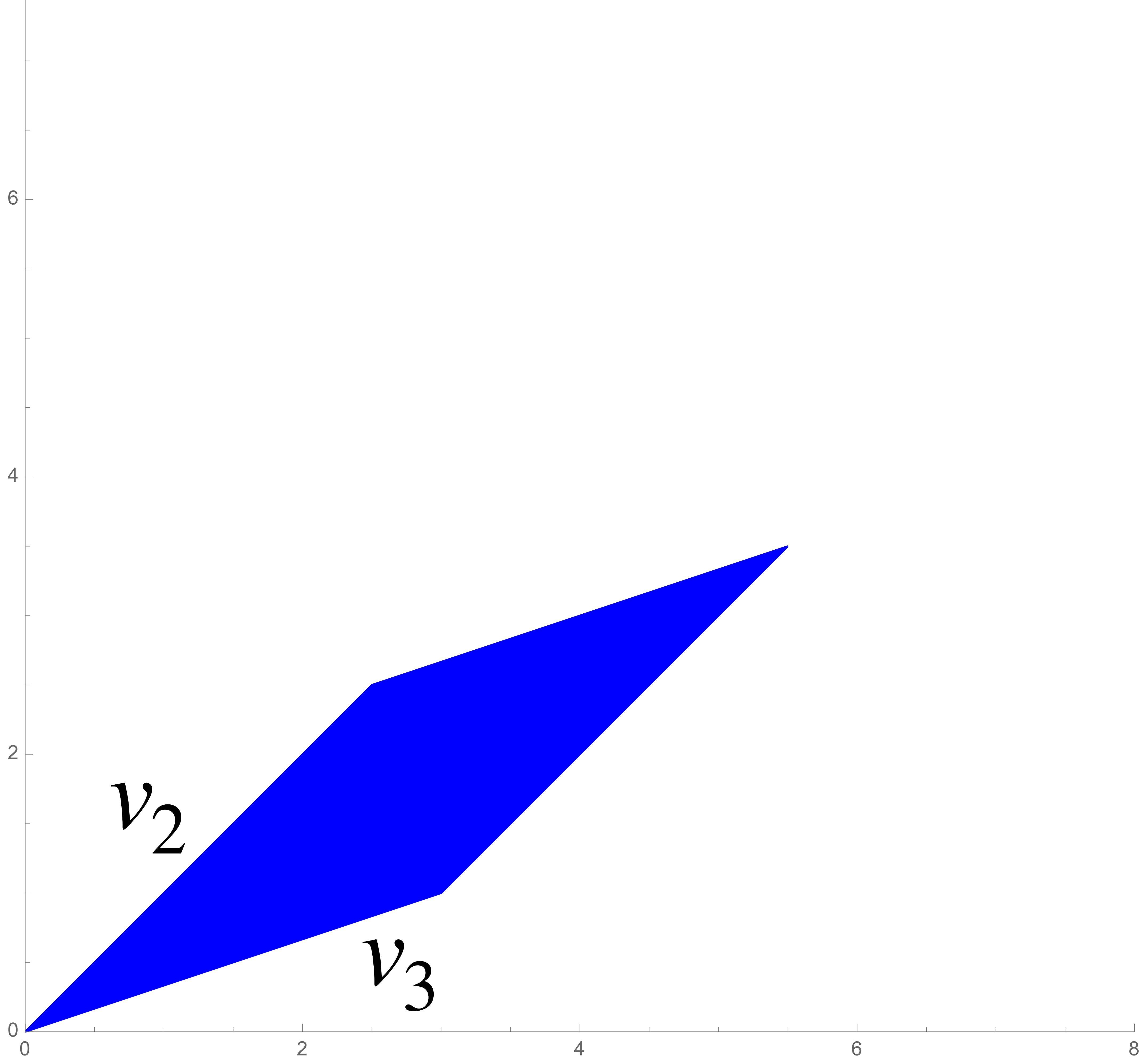

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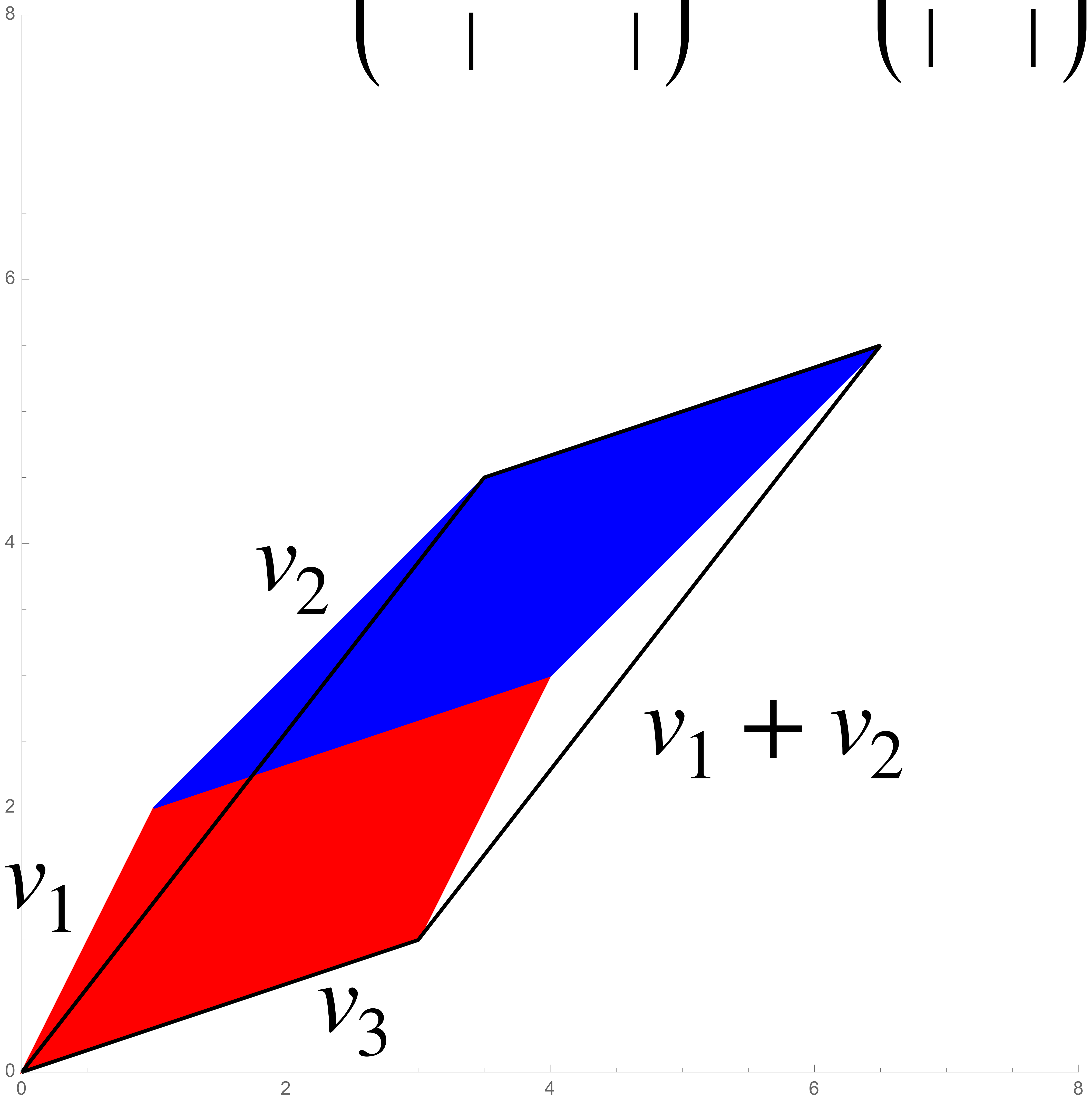

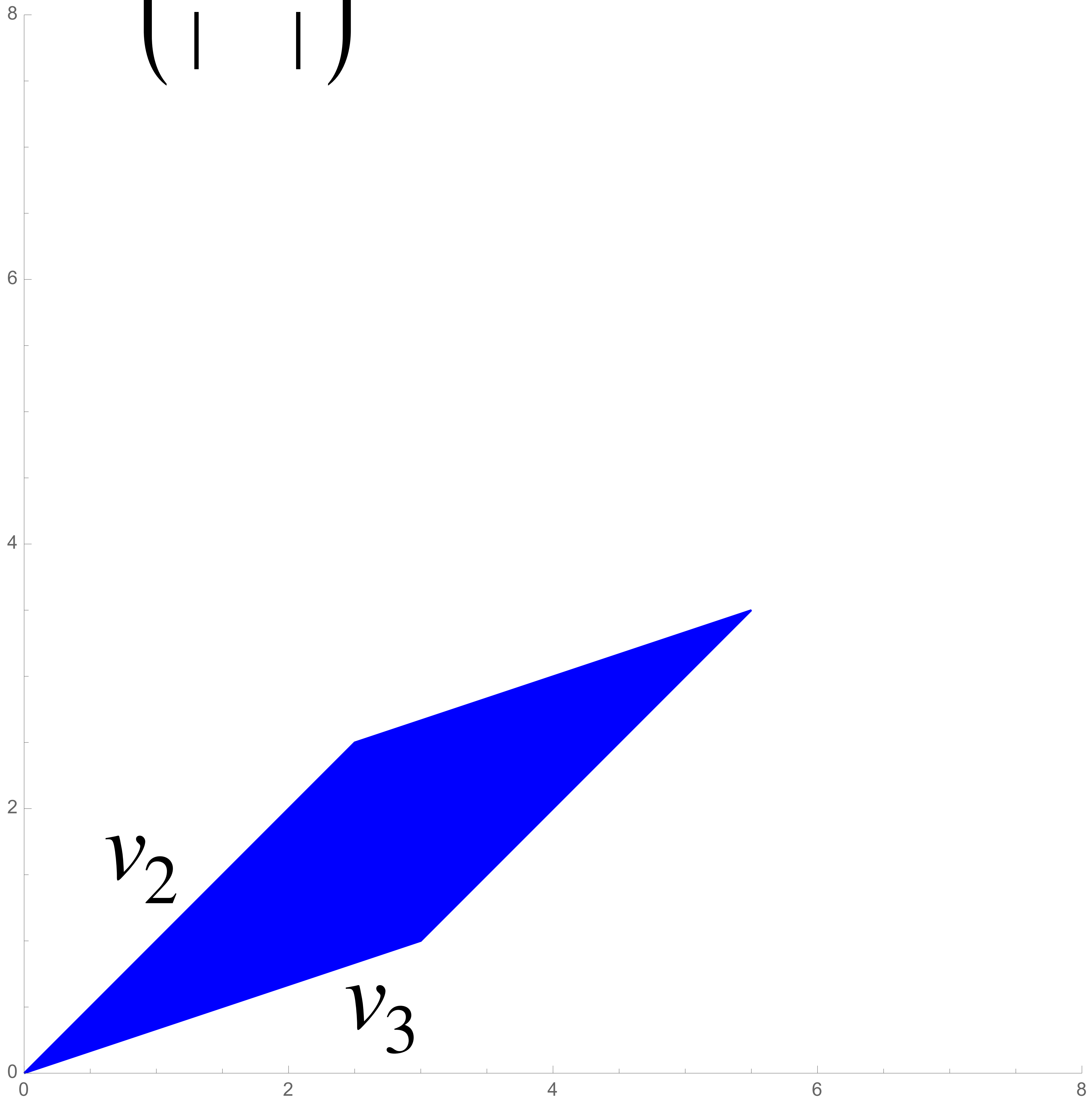

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